

ADV: Statistics (Adv), S1 Probability & Probability Distributions (Adv)
Relative Frequency (Y11)

Teacher: Denis Vlismas

Exam Equivalent Time: 73.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

HISTORICAL CONTRIBUTION

- *S1 Probability* is a Year 11 topic that contains a mixture of old course and new syllabus content.
- The new *Advanced* course in this topic area has a significant overlap with the *Standard 2* syllabus and also introduces some brand new content.
- *S1 Probability* has been split into four sub-categories for the purposes of this analysis: 1-Multi-Stage Events, 2-Relative Frequency, 3-Conditional Probability and Venn Diagrams and 4-Discrete Probability Distributions.
- This analysis looks at *Relative Frequency*.

HSC ANALYSIS - What to expect and common pitfalls

- *Relative Frequency* is a sub-topic that has previously only been examined in the *Standard* course but now represents common content with the new *Advanced* syllabus.
- *Relative Frequency* was not examined in the first two *Advanced* exams in 2020-21.
- It has been a substantial contributor to past *Std2* HSC exams, accounting for a very healthy average of 2.7% per year.
- Our database is made up of an extensive history of past HSC questions from the *Std2* course, which can be identified by the "STD2 S2" in their title.
- Although generally well answered, note the sub-50% mean marks in 2018 Adv 26a, 2017 Adv 29c and in 2016 Adv 23 MC that deserve attention (note that all mean marks and comments are taken from *Std2* results).

Questions

1. Probability, STD2 S2 2013 HSC 7 MC

In an experiment, a standard six-sided die was rolled 72 times. The results are shown in the table.

<i>Number on die</i>	<i>Frequency</i>
1	6
2	12
3	10
4	20
5	9
6	15

Which number on the die was obtained the expected number of times?

- (A) 1
(B) 2
(C) 3
(D) 6
-

2. Probability, STD2 S2 2007 HSC 2 MC

Each student in a class is given a packet of lollies. The teacher records the number of red lollies in each packet using a frequency table.

<i>Number of red lollies in each packet</i>	<i>Frequency</i>
0	2
1	4
2	2
3	7
4	3
5	1

What is the relative frequency of a packet of lollies containing more than three red lollies?

- (A) $\frac{4}{19}$
(B) $\frac{4}{15}$
(C) $\frac{11}{19}$
(D) $\frac{11}{15}$

3. Probability, STD2 S2 2017 HSC 5 MC

In a survey of 200 randomly selected Year 12 students it was found that 180 use social media.

Based on this survey, approximately how many of 75 000 Year 12 students would be expected to use social media?

- A. 60 000
B. 67 500
C. 74 980
D. 75 000

4. Probability, STD2 S2 2009 HSC 8 MC

Some men and women were surveyed at a football game. They were asked which team they supported. The results are shown in the two-way table.

	<i>Team A</i>	<i>Team B</i>	<i>Totals</i>
Men	125	100	225
Women	75	90	165
Totals	200	190	390

What percentage of the women surveyed supported *Team B*, correct to the nearest percent?

- (A) 23%
(B) 45%
(C) 47%
(D) 55%

5. Probability, STD2 S2 2007 HSC 16 MC

Leanne copied a two-way table into her book.

	<i>Male</i>	<i>Female</i>	<i>Totals</i>
<i>Full-time work</i>	279	356	635
<i>Part-time work</i>	187	439	716
<i>Totals</i>	466	885	1351

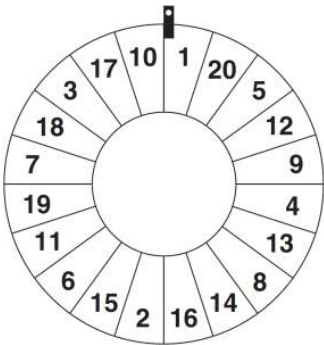
Leanne made an error in copying one of the values in the shaded section of the table.

Which value has been incorrectly copied?

- (A) The number of males in full-time work
(B) The number of males in part-time work
(C) The number of females in full-time work
(D) The number of females in part-time work

6. Probability, STD2 S2 2009 HSC 9 MC

A wheel has the numbers 1 to 20 on it, as shown in the diagram. Each time the wheel is spun, it stops with the marker on one of the numbers.



The wheel is spun 120 times.
How many times would you expect a number less than 6 to be obtained?

- (A) 20
- (B) 24
- (C) 30
- (D) 36

7. Probability, STD2 S2 2010 HSC 12 MC

A group of 347 people was tested for flu and the results were recorded. The flu test results are not always accurate.

	Test results		
	Test indicated flu	Test did not indicate flu	Total
People with flu	72	3	75
People without flu	16	256	272
	88	259	347

A person is selected at random from the tested group.
What is the probability that their test result is accurate, to the nearest per cent?

- (A) 21%
- (B) 22%
- (C) 95%
- (D) 96%

8. Probability, STD2 S2 2013 HSC 10 MC

Students studying vocational education courses were surveyed about their living arrangements.

	Females	Males	Totals
Living with parent(s)	46	155	201
Not living with parent(s)	182	122	304
Totals	228	277	505

One of these students is selected at random.
What is the probability that this student is male and living with his parent(s)?

- (A) 31%
- (B) 40%
- (C) 56%
- (D) 77%

9. Probability, STD2 S2 2005 HSC 16 MC

On a television game show, viewers voted for their favourite contestant. The results were recorded in the two-way table.

	<i>Male viewers</i>	<i>Female viewers</i>
<i>Contestant 1</i>	1372	3915
<i>Contestant 2</i>	2054	3269

One male viewer was selected at random from all of the male viewers.
What is the probability that he voted for Contestant 1?

- (A) $\frac{1372}{10\,610}$
- (B) $\frac{1372}{5287}$
- (C) $\frac{1372}{3426}$
- (D) $\frac{1372}{2054}$

10. Probability, STD2 S2 2006 HSC 6 MC

Marcella is planning to roll a standard six-sided die 60 times.
How many times would she expect to roll the number 4?

- (A) 6
- (B) 10
- (C) 15
- (D) 20

11. Probability, STD2 S2 2018 HSC 20 MC

During a year, the maximum temperature each day was recorded. The results are shown in the table.

	<i>Number of days with maximum temperature less than 25°C</i>	<i>Number of days with maximum temperature greater than or equal to 25°C</i>	<i>Total number of days</i>
Summer	42	48	90
Autumn	57	35	92
Winter	71	21	92
Spring	53	38	91
Total number of days	223	142	365

From the days with a maximum temperature less than 25°C, one day is selected at random.
What is the probability, to the nearest percentage, that the selected day occurred during winter?

- A. 19%
- B. 25%
- C. 32%
- D. 77%

12. Probability, STD2 S2 2012 HSC 17 MC

A spinner with different coloured sectors is spun 40 times. The results are recorded in the table.

<i>Colour obtained</i>	<i>Frequency</i>
Red	2
Yellow	4
Blue	6
Orange	
Green	10
Purple	12

What is the relative frequency of obtaining the colour orange?

- (A) $\frac{3}{20}$
 (B) $\frac{1}{5}$
 (C) 6
 (D) 8

13. Probability, STD2 S2 2016 HSC 23 MC

A group of 485 people was surveyed. The people were asked whether or not they smoke. The results are recorded in the table.

	<i>Smokers</i>	<i>Non-smokers</i>	<i>Total</i>
Male	88	176	264
Female	68	153	221
	156	329	485

A person is selected at random from the group.

What is the approximate probability that the person selected is a smoker OR is male?

- (A) 33%
 (B) 18%
 (C) 68%
 (D) 87%

14. Probability, STD2 S2 2011 HSC 25c

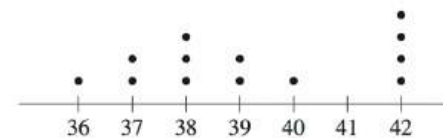
At another school, students who use mobile phones were surveyed. The set of data is shown in the table.

	<i>Pre-paid</i>	<i>Plan</i>	<i>TOTAL</i>
<i>Female students</i>	172	147	319
<i>Male students</i>	158	103	261
<i>TOTAL</i>	330	250	

- i. How many students were surveyed at this school? **(1 mark)**
 ii. Of the female students surveyed, one is chosen at random. What is the probability that she uses pre-paid? **(1 mark)**
 Ten new male students are surveyed and all ten are on a plan. The set of data is updated to include this information.
 iii. What percentage of the male students surveyed are now on a plan? Give your answer to the nearest per cent. **(1 mark)**

15. Probability, STD2 S2 2012 HSC 26e

The dot plot shows the number of push-ups that 13 members of a fitness class can do in one minute.



- i. What is the probability that a member selected at random from the class can do more than 38 push-ups in one minute? **(1 mark)**
 ii. A new member who can do 32 push-ups in one minute joins the class.
 Does the addition of this new member to the class change the probability calculated in part (i)? Justify your answer. **(1 mark)**

16. Probability, STD2 S2 2006 HSC 26c

A new test has been developed for determining whether or not people are carriers of the Gaussian virus.

Two hundred people are tested. A two-way table is being used to record the results.

	Positive	Negative
Carrier	74	12
Not a carrier	16	A

- i. What is the value of **A**? (1 mark)
- ii. A person selected from the tested group is a carrier of the virus.
What is the probability that the test results would show this? (2 marks)
- iii. For how many of the people tested were their test results inaccurate? (1 mark)

17. Probability, STD2 S2 2008 HSC 26a

Cecil invited 175 movie critics to preview his new movie. After seeing the movie, he conducted a survey. Cecil has almost completed the two-way table.

	Aged <40	Aged ≥40	Totals
Movie critics who liked the movie	65		102
Movie critics who did not like the movie		31	
Totals		A	175

- i. Determine the value of **A**. (1 mark)
- ii. A movie critic is selected at random.
What is the probability that the critic was less than 40 years old and did not like the movie? (2 marks)
- iii. Cecil believes that his movie will be a box office success if 65% of the critics who were surveyed liked the movie.
Will this movie be considered a box office success? Justify your answer. (1 mark)

18. Probability, STD2 S2 2015 HSC 26e

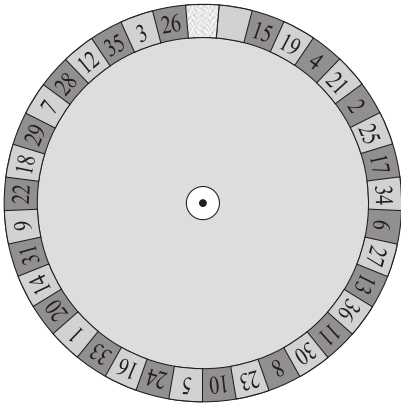
The table shows the relative frequency of selecting each of the different coloured jelly beans from packets containing green, yellow, black, red and white jelly beans.

Colour	Relative frequency
Green	0.32
Yellow	0.13
Black	0.14
Red	
White	0.24

- i. What is the relative frequency of selecting a red jelly bean? (1 mark)
- ii. Based on this table of relative frequencies, what is the probability of NOT selecting a black jelly bean? (1 mark)

19. Probability, STD2 S2 2019 HSC 20

A roulette wheel has the numbers 0, 1, 2, ..., 36 where each of the 37 numbers is equally likely to be spun.



If the wheel is spun 18 500 times, calculate the expected frequency of spinning the number 8. (2 marks)

20. Probability, STD2 S2 2011 HSC 24b

A die was rolled 72 times. The results for this experiment are shown in the table.

<i>Number obtained</i>	<i>Frequency</i>
1	16
2	11
3	A
4	8
5	12
6	15

- Find the value of **A**. (1 mark)
- What was the relative frequency of obtaining a 4. (1 mark)
- If the die was unbiased, which number was obtained the expected number of times? (1 mark)

21. Probability, STD2 S2 2004 HSC 25c

Lie detector tests are not always accurate. A lie detector test was administered to 200 people.

The results were:

- 50 people lied. Of these, the test indicated that 40 had lied;
- 150 people did NOT lie. Of these, the test indicated that 20 had lied.

- Copy the table into your writing booklet and complete it using the information above

	<i>Test indicated a lie</i>	<i>Test did not indicate a lie</i>	<i>Total</i>
People who lied			50
People who did NOT lie			150

- For how many of the people tested was the lie detector test accurate?
- For what percentage of the people tested was the test accurate?
- What is the probability that the test indicated a lie for a person who did NOT lie?

22. Probability, STD2 S2 2008 HSC 26b

The retirement ages of two million people are displayed in a table.

<i>Retirement age</i>	<i>Number of people (thousands)</i>
36–40	5
41–45	10
46–50	20
51–55	35
56–60	180
61–65	700
66–70	500
71–75	400
76–80	150

- What is the relative frequency of the 51–55 year retirement age? (1 mark)
- Describe the distribution. (1 mark)

23. Probability, STD2 S2 2017 HSC 29c

A group of Year 12 students was surveyed. The students were asked whether they live in the city or the country. They were also asked if they have ever waterskied.

The results are recorded in the table.

	Have waterskied	Have never waterskied
Live in the city	150	2500
Live in the country	70	800

- A person is selected at random from the group surveyed. Calculate the probability that the person lives in the city and has never waterskied. (2 marks)
- A newspaper article claimed that Year 12 students who live in the country are more likely to have waterskied than those who live in the city.
Is this true, based on the survey results? Justify your answer with relevant calculations. (2 marks)

24. Probability, STD2 S2 2018 HSC 26a

Jeremy rolled a biased 6-sided die a number of times. He recorded the results in a table.

Number	1	2	3	4	5	6
Frequency	23	19	48	20	21	19

What is the relative frequency of rolling a 3? (1 mark)

25. Probability, STD2 S2 2009 HSC 28d

In an experiment, two unbiased dice, with faces numbered 1, 2, 3, 4, 5, 6 are rolled 18 times.

The difference between the numbers on their uppermost faces is recorded each time. Juan performs this experiment twice and his results are shown in the tables.

Experiment 1

Difference	Frequency
0	3
1	3
2	2
3	4
4	3
5	3

Experiment 2

Difference	Frequency
0	4
1	4
2	3
3	3
4	2
5	2

Juan states that Experiment 2 has given results that are closer to what he expected than the results given by Experiment 1.

Is he correct? Explain your answer by finding the sample space for the dice differences and using theoretical probability. (4 marks)

Worked Solutions

1. Probability, STD2 S2 2013 HSC 7 MC

$$\text{Probability of rolling a specific number} = \frac{1}{6}$$

$\therefore$ After 72 rolls, a specific number is expected

$$\frac{1}{6} \times 72 = 12 \text{ times.}$$

$\Rightarrow B$

2. Probability, STD2 S2 2007 HSC 2 MC

Packets with more than 3

$$= 3 + 1 = 4$$

Total packets = 19

$$\therefore \text{Relative Frequency} = \frac{4}{19}$$

$\Rightarrow A$

3. Probability, STD2 S2 2017 HSC 5 MC

$$\begin{aligned} \text{Expected number} &= \frac{180}{200} \times 75\,000 \\ &= 67\,500 \end{aligned}$$

$\Rightarrow B$

4. Probability, STD2 S2 2009 HSC 8 MC

$$\begin{aligned} \% \text{ Women for Team B} &= \frac{\# \text{ Women for Team B}}{\# \text{ Women surveyed}} \\ &= \frac{90}{165} \\ &= 54.54\ldots \% \end{aligned}$$

$\Rightarrow D$

5. Probability, STD2 S2 2007 HSC 16 MC

By checking row and column total, the number of females part-time work is incorrect

$\Rightarrow D$

6. Probability, STD2 S2 2009 HSC 9 MC

$$P(\text{number} < 6) = \frac{5}{20} = \frac{1}{4}$$

$$\begin{aligned}\therefore \text{Expected times} &= \frac{1}{4} \times \text{times spun} \\ &= \frac{1}{4} \times 120 \\ &= 30\end{aligned}$$

$\Rightarrow C$

7. Probability, STD2 S2 2010 HSC 12 MC

$$\begin{aligned}P(\text{Test accurate}) &= \frac{\text{Accurate readings}}{\text{Total tested}} \\ &= \frac{72 + 256}{347} \\ &= 94.5244\ldots \%\end{aligned}$$

$\Rightarrow C$

8. Probability, STD2 S2 2013 HSC 10 MC

Males living with their parents is 155.

$$\therefore P = \frac{155}{505} = 0.30693\ldots \%$$

$\Rightarrow A$

9. Probability, STD2 S2 2005 HSC 16 MC

$$\begin{aligned}\text{Total male viewers} &= 1372 + 2054 \\ &= 3426\end{aligned}$$

P (Male viewer chosen voted for C1)

$$\begin{aligned}&= \frac{\text{Males who voted for C1}}{\text{Total male viewers}} \\ &= \frac{1372}{3426}\end{aligned}$$

$\Rightarrow C$

10. Probability, STD2 S2 2006 HSC 6 MC

$$P(4) = \frac{1}{6}$$

$\therefore$ Expected times to roll 4

$$\begin{aligned}&= \frac{1}{6} \times \text{number of rolls} \\ &= \frac{1}{6} \times 60 \\ &= 10\end{aligned}$$

$\Rightarrow B$

11. Probability, STD2 S2 2018 HSC 20 MC

$$\begin{aligned}P(\text{winter day}) &= \frac{\text{winter days} < 25}{\text{total days} < 25} \times 100 \\ &= \frac{71}{223} \times 100 \\ &= 31.8\ldots \%\end{aligned}$$

$\Rightarrow C$

12. Probability, STD2 S2 2012 HSC 17 MC

Total frequency = 40 spins

$$\begin{aligned}\text{Orange freq.} &= 40 - (2 + 4 + 6 + 10 + 12) \\ &= 6\end{aligned}$$

$$\therefore \text{Relative freq.} = \frac{6}{40} = \frac{3}{20}$$

$\Rightarrow A$

♦♦ Mean mark 34%
COMMENT: Note that relative frequency is the frequency of an event *divided* by the total frequencies (i.e. the probability).

13. Probability, STD2 S2 2016 HSC 23 MC

P (Smoker or a male)

$$= \frac{\text{Total males} + \text{female smokers}}{\text{Total surveyed}}$$

$$= \frac{264 + 68}{485}$$

$$= 0.684\ldots$$

$\Rightarrow C$

♦♦ Mean mark 34%.

14. Probability, STD2 S2 2011 HSC 25c

i. $\# \text{ Students surveyed} = 319 + 261 = 580$

ii. $P(\text{Female uses prepaid}) = \frac{\# \text{ Females on prepaid}}{\text{Total females}}$
 $= \frac{172}{319}$
 $= 0.53918\dots$
 $= 54\% \text{ (nearest \%)}$

iii. $\% \text{ Males on plan} = \frac{\# \text{ Males on plan} + 10}{\text{Total males} + 10}$
 $= \frac{103 + 10}{261 + 10}$
 $= \frac{113}{271}$
 $= 0.4169\dots$
 $= 42\% \text{ (nearest \%)}$

15. Probability, STD2 S2 2012 HSC 26e

i. $P = \frac{\# \text{ Members} > 38 \text{ push-ups}}{\text{Total members}}$
 $= \frac{7}{13}$

ii. Yes.

$$P(+ \text{ New member}) = \frac{\text{Members} > 38 \text{ push-ups}}{\text{Total members}}$$
$$= \frac{7}{14} \neq \frac{7}{13}$$

MARKER'S COMMENT: The most successful candidates in this part used the fraction $\frac{7}{14}$ in their answer rather than relying solely on words in their explanation.

16. Probability, STD2 S2 2006 HSC 26c

i. $A = 200 - (74 + 12 + 16)$
 $= 98$

ii. $P = \frac{\# \text{ Positive carriers}}{\text{Total carriers}}$
 $= \frac{74}{86}$
 $= \frac{37}{43}$

iii. $\# \text{ People with inaccurate results}$
 $= 12 + 16$
 $= 28$

17. Probability, STD2 S2 2008 HSC 26a

i. Critics liked and ≥ 40

$$= 102 - 65$$

$$= 37$$

$$\therefore A = 37 + 31 = 68$$

ii. Critics did not like and < 40

$$= 175 - 65 - 37 - 31$$

$$= 42$$

$$\therefore P(\text{not like and } < 40)$$

$$= \frac{42}{175}$$

$$= \frac{6}{25}$$

iii. Critics liked = 102

$$\% \text{ Critics liked} = \frac{102}{175} \times 100$$

$$= 58.28\ldots \%$$

$\therefore$ Movie NOT a box office success,
as $< 65\%$ of critics liked it.

18. Probability, STD2 S2 2015 HSC 26e

i. Relative frequency of red

$$= 1 - (0.32 + 0.13 + 0.14 + 0.24)$$

$$= 1 - 0.83$$

$$= 0.17$$

ii. $P(\text{not selecting black})$

$$= 1 - P(\text{selecting black})$$

$$= 1 - 0.14$$

$$= 0.86$$

19. Probability, STD2 S2 2019 HSC 20

$$P(8) = \frac{1}{37}$$

$\therefore$ Expected Frequency (8)

$$= \frac{1}{37} \times 18\,500$$

$$= 500$$

20. Probability, STD2 S2 2011 HSC 24b

i. Since die rolled 72 times

$$\therefore A = 72 - (16 + 11 + 8 + 12 + 15)$$

$$= 72 - 62$$

$$= 10$$

ii. Relative frequency of 4 = $\frac{8}{72}$

$$= \frac{1}{9}$$

♦ Mean mark 38%
IMPORTANT: Many students confused 'relative frequency' with 'frequency' and incorrectly answered 8.

iii. Expected frequency of any number

$$= \frac{1}{6} \times 72$$

$$= 12$$

$\therefore$ 5 was obtained the expected number of times.

21. Probability, STD2 S2 2004 HSC 25c

i.

	<i>Test indicated a lie</i>	<i>Test did not indicate a lie</i>	<i>Total</i>
People who lied	40	10	50
People who did NOT lie	20	130	150

ii. # Accurate readings

$$= 40 + 130$$

$$= 170$$

iii. Percentage of people with accurate readings

$$= \frac{\# \text{ Accurate readings}}{\text{Total readings}} \times 100$$

$$= \frac{170}{200}$$

$$= 85\%$$

iv. P(lie detected when NOT a lie)

$$= \frac{20}{150}$$

$$= \frac{2}{15}$$

22. Probability, STD2 S2 2008 HSC 26b

i. Relative frequency (51 – 55)

$$= \frac{\# \text{ People (51-55)}}{\text{Total People}}$$

$$= \frac{35\,000}{2\,000\,000}$$

$$= \frac{7}{400}$$

ii. Distribution is negatively skewed because
as age increases, so does the number of
people in each age bracket.

23. Probability, STD2 S2 2017 HSC 29c

i. $P = \frac{\text{live in city, not skied}}{\text{total surveyed}}$

$$= \frac{2500}{3520}$$

$$= \frac{125}{176}$$

♦ Mean mark part (ii) 47%.

ii. $P(\text{live in country, skied}) = \frac{70}{(70 + 800)}$

$$= 0.0804\dots$$

$$= 8\%$$

$$P(\text{live in city, skied}) = \frac{150}{(150 + 2500)}$$

$$= 0.0566$$

$$= 6\%$$

Since $8\% > 6\%$, the statement is true.

24. Probability, STD2 S2 2018 HSC 26a

$$\text{Rel Freq} = \frac{\text{number of 3's rolled}}{\text{total rolls}}$$

$$= \frac{48}{150}$$

$$= \frac{8}{25}$$

♦ Mean mark 40%.

25. Probability, STD2 S2 2009 HSC 28d

Sample space for dice differences

	1	2	3	4	5	6
1	0	1	2	3	4	5
2	1	0	1	2	3	4
3	2	1	0	1	2	3
4	3	2	1	0	1	2
5	4	3	2	1	0	1
6	5	4	3	2	1	0

♦♦♦ Mean mark 7%. Toughest question in the 2009 exam.
MARKER'S COMMENT: This question guides students by asking for an explanation using the *sample space* for the dice differences. This step alone received 2 full marks. Note that instructions to **explain your answer** requires mathematical calculations to support an argument.

Differences	Probabilities	Expected frequencies
0	$\frac{6}{36}$	3
1	$\frac{10}{36}$	5
2	$\frac{8}{36}$	4
3	$\frac{6}{36}$	3
4	$\frac{4}{36}$	2
5	$\frac{2}{36}$	1

Differences from expected frequency	
Experiment 1	Experiment 2
0	1
2	1
2	1
1	0
1	0
2	1
8	4

Juan is correct. The table shows Experiment 1 has greater total differences to the expected frequencies than Experiment 2